Reflections in the plane

Four mirrors, four matrices. The first three are special cases of the fourth — substituting $\alpha = 0$, $\frac{\pi}{2}$ and $\frac{\pi}{4}$ into the general form recovers them exactly.

4
MIRRORS

MIRRORS ALONG THE AXES — 2



The x-axis | $\alpha = 0$

MATRIX	EFFECT ON (X, Y)	FIXED DIRECTION
$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	$(x, y) \mapsto (x, -y)$	the x-axis

Diagonal, so its action is entrywise: the first coordinate survives and the second flips. The eigenvalues sit on the diagonal, $+1$ and -1, exactly as the general result predicts.

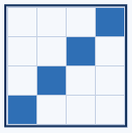


The y-axis | $\alpha = \frac{\pi}{2}$

MATRIX	EFFECT ON (X, Y)	FIXED DIRECTION
$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$	$(x, y) \mapsto (-x, y)$	the y-axis

The same matrix with the signs exchanged, which makes it the negative of the row above. Negating both diagonal entries of I instead gives $-I$, a rotation by π, which reverses both coordinates and has $\det = +1$.

MIRROR AT 45 DEGREES — 1

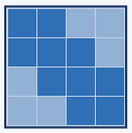


The line $y = x$ | $\alpha = \frac{\pi}{4}$

MATRIX	EFFECT ON (X, Y)	FIXED DIRECTION
$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$(x, y) \mapsto (y, x)$	the line $y = x$

Swapping coordinates is a reflection, which is worth noticing because it also describes the **transpose** at the level of a 2×2 permutation. Its own inverse, as every reflection is.

THE GENERAL MIRROR — 1



Line at angle α | any line through the origin

MATRIX	EFFECT ON (X, Y)	FIXED DIRECTION
$\begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{bmatrix}$	reflects across that line	the mirror line itself

The 2α is not a typo: rotating the mirror by α turns the image by 2α, since the incoming and outgoing rays each make angle α with it. Symmetric and orthogonal at once, which is unusual and is why $A^{-1} = A^{\mathsf{T}} = A$.

Every reflection has $\det = -1$, eigenvalues $+1$ along the mirror and -1 across it, and equals its own inverse. Those three facts hold whatever α is, which is what makes reflection a **family** rather than a list — and why the general row is the only one that has to be remembered.